

1. What is the difference between readyState and status in XMLHttpRequest?

Property:

readyState

status

Description:

Shows the current state or phase of the request. It changes as the request is processed.

Shows the HTTP response code from the server (e.g., 200 = OK, 404 = Not Found).

Example:

```
xhr.readyState // tells what phase the request is in (0–4)
```

```
xhr.status      // tells if the request was successful (like 200)
```

readyState values:

Value: 0 1 2 3 4

Meaning:

Request not initialized

Connection established

Request received

Processing request

Request finished and response is ready

2. How can you make the request asynchronous?

You can make a request asynchronous by setting the third parameter to true in the open() method of XMLHttpRequest.

Example:

```
xhr.open("GET", "data.json", true); // true = asynchronous
```

```
xhr.send();
```

If you use false, the request becomes synchronous (which blocks the browser until the response is complete).

3. Why is asynchronous communication recommended for web applications?

Asynchronous communication is recommended because it:

Does not block the user interface — users can still interact with the page while data is being loaded.

Improves performance — only specific parts of the page are updated without reloading the whole page.

Enhances user experience — responses appear smoother and faster.

Allows background processing — requests can run quietly while users continue browsing.

Example:

When you send a form or load content dynamically (like chat messages or notifications), the page doesn't refresh — thanks to asynchronous communication.